L2a: Are Risk-Free Assets Really Risk-Free?

CHEME 5660 Cornell University

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we introduce U.S. marketable Treasury securities, build their cash-flow and pricing conventions, and ask what risk-free does and does not mean.

Today's concepts

- Describe Treasury securities: distinguish bills, notes, and bonds, why the government issues them, and what makes them attractive.
- Value Treasury cash flows: represent each security as dated cash flows, and connect market prices to discount factors and quoted yields.
- Evaluate the risk-free label: why Treasuries serve as a reference point, and what inflation, reinvestment, liquidity, and mark-to-market risks remain.

Companion notes: L2a lecture notes and notebooks

Lectures and Examples

Lecture: CHEME-5660-L2a-Lecture-TreasurySecurities-Fall-2026.ipynb

We compute the price of Treasury securities at auction, then compare the computed price with the actual auction price.

Example: CHEME-5660-L2a-TBills-Pricing-Fall-2026-WorkedExample.ipynb

Example: CHEME-5660-L2a-NotesBonds-Pricing-Fall-2026-WorkedExample.ipynb

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Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

Quick review before we start: the time value of money, abstract assets, and net present value.

Review: Yields and Accumulation Factors

Dated dollars have different units. A dollar today and a dollar at a future date cannot be added until a stated rule expresses them at a common valuation date.

Let P be an amount held today, let y be a quoted nominal annual yield, and let n be the number of compounding intervals per year. At period index j, the equivalent future amount is given by:

$$F_j = \underbrace{(1 + y/n)^j}_{\mathcal{D}_{j,0}(y;n)} P.$$

$\mathcal{D}_{j,0}(y;n)$ converts time-0 dollars into period-j dollars and its inverse converts them back. The continuously compounded equivalent is $g_y := n \log(1 + y/n)$, so $\mathcal{D}_{T,0}(y;n) = (1 + y/n)^{nT} = e^{g_y T}$ over T years.

Careful: g is a growth rate and $r = (\Delta t)\, g$ its log return. Convert a quoted y before comparing.

Review: Abstract Assets and Net Present Value

Abstract assets describe financial value as dated signed cash flows. Because dated dollars have different units, each cash flow must be converted to a common valuation date before the values are added.

Let N be the final cash-flow index and let $\bar{c}_j$ be the signed net cash flow at index j. Under a constant nominal yield y, the time-0 net present value is given by:

$$\mathrm{NPV}_0(y) = \sum_{j=0}^{N} \mathcal{D}_{j,0}^{-1}(y;n)\,\bar{c}_j.$$

Every term in the sum is measured in time-0 dollars.

Today: set $\mathrm{NPV}_0 = 0$ and solve for the price. That is what pricing a Treasury security means.

Treasury Bills

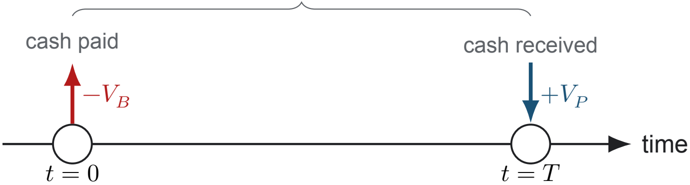
A U.S. Treasury bill is a short-term zero-coupon debt security issued by the Department of the Treasury, with maturities running from a few days to one year. Bills are sold below face value and pay no periodic interest.

Writing V_B for the purchase price and V_P for the face value, the difference $V_P - V_B$ is a nominal dollar gain, not a return: it is a dollar amount, and it says nothing about the rate until a horizon and a convention are attached to it.

Where to buy: TreasuryDirect, or a bank or broker. Bills may be held to maturity or sold in the secondary market.

Treasury Bill Cash Flows

zero-coupon bill: one payment, one receipt



$$V_B = \mathcal{D}_{T,0}^{-1}(y;n)\, V_P = e^{-g_y T} V_P$$

dollar gain: $V_P - V_B$; there are no coupons

$$g_B = T^{-1} \log(V_P / V_B), \quad r_B = T g_B$$

Valuing a T-Bill

A bill has two cash-flow events for the investor: the purchase price $-V_B$ at time 0, and the face-value receipt $+V_P$ at maturity $T > 0$ years. Let y be a nominal annual yield with frequency n, and $g_y = n\log(1 + y/n)$. Setting $\mathrm{NPV}_0(y) = 0$ and solving for the price gives:

$$V_B = \mathcal{D}_{T,0}^{-1}(y;n)\,V_P = \left(1 + \frac{y}{n}\right)^{-nT} V_P = e^{-g_y T} V_P.$$

Its holding-period growth rate and log return are given by:

$$g_B = \frac{1}{T}\log\left(\frac{V_P}{V_B}\right), \qquad r_B = T\,g_B, \qquad \text{and here } g_B = g_y.$$

Careful: quoted bank-discount and investment rates are *not* automatically y or g_y. Reconcile conventions first.

Treasury Notes and Bonds

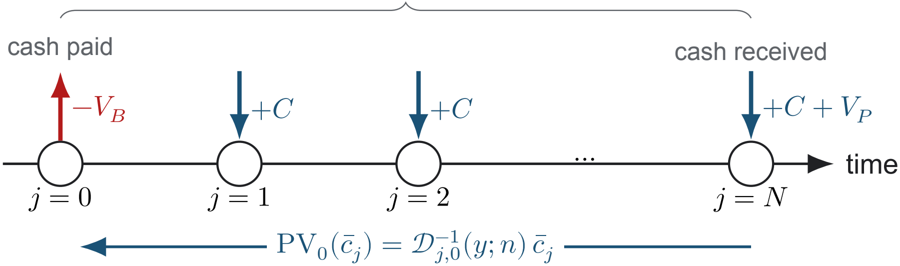
Treasury notes and bonds pay semiannual coupons and repay face value at maturity. They share one cash-flow structure and differ in their standard maturity ranges.

Let $T > 0$ be maturity in years, let n be the coupon frequency per year, and let $N = nT$ be the number of remaining coupon dates. The annual coupon rate is c, so each coupon pays $C = (c/n)V_P$. From the investor's perspective the dated net cash flows are:

$$\bar{c}_0 = -V_B, \qquad \bar{c}_j = C \ \ (j = 1, \dots, N-1), \qquad \bar{c}_N = C + V_P.$$

Note and Bond Cash Flows

coupon security: $N = nT$ **coupons, then par at maturity**



net cash flow: $\bar{c}_0 = -V_B$, $\bar{c}_j = C$ for $0 < j < N$, $\bar{c}_N = C + V_P$

Valuing Notes and Bonds

Let y be the constant nominal annual yield used for valuation. Setting the net present value to zero and solving for the price gives:

$$0 = \underbrace{-V_B}_{\text{cost}} + \underbrace{\mathcal{D}_{N,0}^{-1}(y;n)V_P}_{\text{par payment}} + \underbrace{C\sum_{j=1}^{N}\mathcal{D}_{j,0}^{-1}(y;n)}_{\text{coupon payments}}$$

$$V_B = \mathcal{D}_{N,0}^{-1}(y;n)V_P + C\sum_{j=1}^{N}\mathcal{D}_{j,0}^{-1}(y;n), \qquad \mathcal{D}_{j,0}^{-1}(y;n) = \left(1+\frac{y}{n}\right)^{-j}.$$

The nominal yield y is a quotation and valuation convention. Its continuously compounded equivalent is $g_y = n\log(1+y/n)$, but a coupon security's realized growth also depends on what the coupons earn once received, and on any sale price before maturity.

Are Treasury Securities Risk-Free?

Not completely. They are called risk-free because the chance of a missed promised nominal-dollar payment is low. That is not a guarantee of profit over every holding period.

Two risks that matter here

- Default risk: the issuer may fail to make a promised payment. This is low for U.S. Treasury securities, but it is not equally low for every sovereign issuer.
- Interest-rate risk: if market yields change, the security's price changes. An investor who sells before maturity may realize a gain or a loss.

Case Study: Long-Term Capital Management

LTCM was a fixed-income fund founded in 1994 by Nobel laureates and experienced traders. At its peak it held over $1 trillion in positions, borrowing roughly $25 for every $1 of its own capital.

- Returns after fees ran about 21%, 43%, and 41% over its first three years.
- In 1998 it lost roughly $4.6 billion in four months, through exposure to the Asian and Russian financial crises.
- The Federal Reserve Bank of New York organized a $3.6 billion rescue by a group of banks to keep the losses from spreading.

Its positions assumed certain government-bond prices would stay in line. When Russia defaulted, investors fled to the safest securities and those relationships broke.

Careful: the fund modeled prices, but not what happens to sovereign credit in a crisis.

Case Study: The 2008 Financial Crisis

A second episode, closer to home, in which assumptions about safety and default were tested across the whole financial system.

Overview: a short introduction to the 2008 crisis

Deeper dive: the Financial Crisis Inquiry Commission Report

Summary

This lecture introduced the cash-flow structures, valuation, and risks of U.S. Treasury securities.

Key takeaways

- **Cash flows determine value.** A bill has one payment at maturity, while notes and bonds pay coupons and return face value. In each case the price follows from converting every cash flow to a common valuation date and setting the net present value to zero.
- **Rate conventions matter.** Yield quotations, compounding frequency, and day-count conventions must be reconciled before rates or prices are compared.
- **Risk-free does not mean price-risk free.** Treasury securities carry low nominal default risk, but their market prices move when yields move, so selling before maturity may produce a gain or a loss.

Treasury securities connect dated cash flows, discounting, and market risk.

That’s a wrap. What are some of the interesting things we discussed today?